

Validation of the Fast Calorimeter Simulation in LHCb Fixed-Target Collisions

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ABSTRACT

Monte-Carlo simulations are essential for LHCb physics analyses, but a full detector simulation is computationally expensive, especially for large simulated samples needed for physics analyses after the LHCb experiment upgrade. Fast simulation provides a possible solution, but its performance must be validated carefully before it can be used in physics analyses. This report studies the performance of the fast simulation of the LHCb calorimeter in fixed-target collisions relevant for the reconstruction of $\chi_c \rightarrow (J/\psi \rightarrow \mu^+ \mu^-) \gamma$ decays. Fast-simulation samples are compared with detailed-simulation samples produced for 2024 fixed-target collision systems at $\sqrt{s_{NN}} = 70.9$ GeV, including pH_2 , ppH_2 , pAr , $ppAr$, and $PbAr$ collisions. The validation focuses on photon kinematic and angular variables, calorimeter-region dependence, truth-matching effects, and the resolution of the reconstructed χ_c mass. The results show that the fast simulation reproduces the overall shapes of several relevant distributions, for example the χ_c mass. However, visible discrepancies are observed for low-energy photons, some angular variables, and truth-matched samples. The comparison across collision systems shows that similar patterns are found. Overall, the fast calorimeter simulation provides a reasonable description of the main photon-related observables, while further quantitative studies are needed before its use in precision χ_c production analyses in SMOG2 data. This validation work provides useful feedback for the development of the LHCb fast simulation, and future needs of the simulation samples involving photons in the LHCb fixed-target collisions.

Keywords: LHCb, SMOG2, fixed-target collisions, full simulation, fast calorimeter simulation, photon reconstruction, χ_c production

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1. INTRODUCTION

Monte Carlo simulations are an essential component of modern particle physics analyses [1]. They are used to model the production and decay of particles, their propagation through the detector, and the response of the detector subsystems. In LHCb, simulated samples are required both for the interpretation of existing measurements and for the preparation of future analyses. However, the increasing luminosity of the upgraded LHCb experiment [2] increase the computational cost of producing sufficiently large simulated samples. This makes the development and validation of fast simulation techniques an important part of the LHCb physics programme.

A particularly important contribution to the simulation cost comes from the calorimeter system, using more than 50% of the CPU time [3]. In the standard detailed simulation, the interactions of particles with the detector material are modeled using the GEANT4 [4] toolkit, which provides an accurate description of the detector response but requires a large amount of computing time. To reduce this cost, a fast calorimeter simulation [3] has been developed based on a point-library approach, in which pre-simulated calorimeter responses are reused to approximate the detector response of particles entering the calorimeter. Since this approximation replaces part of the full GEANT4-based simulation, its performance must be carefully validated before it can be used in physics analyses.

The fast calorimeter simulation has already been validated in proton-proton collisions at $\sqrt{s} = 13$ TeV. In addition to the standard proton-proton collision, LHCb can operate the fixed-target collisions using the SMOG2 system, where the circulating LHC beam collides with gas targets at rest. The fast simulation has not yet been validated for fixed-target collisions at $\sqrt{s_{NN}} = 70.9$ GeV. This study is important because the collision energy and geometry are very different, thus the produced particles can have different kinematic distributions. One of the physics motivations of this programme is the study of charmonium production at low collision energies. Charmonium productions are the particles containing a charm quark (c) and a anti-charm quark ($\bar{c}$), including J/ψ , $\psi(2S)$, χ_c states. Usually, the J/ψ and $\psi(2S)$ is decaying into two muon pairs, while χ_c decays to J/ψ and emits a photon. The decay channel of χ_c is:

$$\chi_c \rightarrow (J/\psi \rightarrow \mu^+ \mu^-) \gamma \quad [5]$$

This decay channel is of interest for future SMOG2 analyses. Since the reconstruction of this channel depends directly on the photon measured in the electromagnetic calorimeter, validating the photon performance of the fast calorimeter simulation is necessary before using it for such studies.

The goal of this work is therefore to compare fast-simulation samples with detailed simulation samples for $\chi_c \rightarrow (J/\psi \rightarrow \mu^+ \mu^-) \gamma$ decays in the LHCb fixed-target collision systems. The comparison is performed using simulated 2024 samples at $\sqrt{s_{NN}} = 70.9$ GeV, including the collision systems: pH_2 , ppH_2 , pAr , $ppAr$, and $PbAr$. The validation is mainly based on photon-related variables, such as momentum, transverse momentum, angular variables, pseudorapidity, and calorimeter region. In addition, the reconstructed χ_c mass is studied, since it is the main observable reflected by the photon reconstruction resolution in this decay channel.

The report is organized as follows. First, the LHCb detector, the SMOG2 fixed-target programme, and the calorimeter system are introduced. The simulation samples and the variables used in the validation are then described, including the truth-matching criteria used to select correctly reconstructed signal candidates. The fast and detailed simulations are first compared in the baseline pH_2 collision system, where the dependence on photon kinematics, photon energy, calorimeter region, and truth matching is investigated. The study is then extended to the other fixed-target collision systems in order to evaluate whether the observed differences between fast and detailed simulation are stable across different collision environments.

2. EXPERIMENTAL SETUP

2.1 The LHC and LHCb experiment

The Large Hadron Collider (LHC) [6] is the world’s largest and most powerful particle accelerator that pushes protons or ions to near the speed of light. It first started up on 10 September 2008, and remains the latest addition to CERN’s accelerator complex. The LHC consists of a 27-kilometer ring of superconducting magnets with a number of accelerating structures to boost the energy of the particles along the way. The accelerator sits in a tunnel 100 meters underground at CERN, the European Organization for Nuclear Research, on the Franco-Swiss border near Geneva in Switzerland, as illustrate in Figure 1.

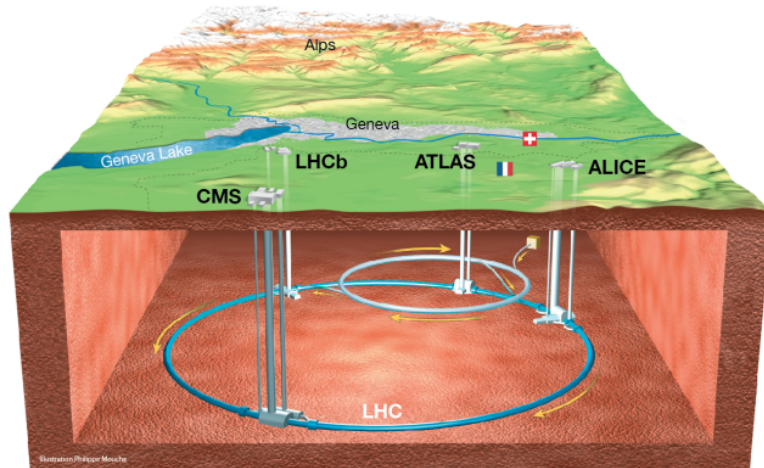


Figure 1: Overall view of the LHC

Inside the accelerator, two high-energy particle beams travel at close to the speed of light before they are made to collide. The beams travel in opposite directions in separate beam pipes. They are guided around the accelerator ring by a strong magnetic field maintained by superconducting electromagnets. The electromagnets are built from coils of special electric cable that operates in a superconducting state, efficiently conducting electricity without resistance or loss of energy. This requires chilling the magnets to $-271.3\text{ }^{\circ}\text{C}$. For this reason, much of the accelerator is connected to a distribution system of liquid helium, which cools the magnets, as well as to other supply services. Thousands of magnets of different varieties and sizes are used to direct the beams around the accelerator. These include 1232 dipole magnets, 15 meters in length, which bend the beams, and 392 quadrupole magnets, each 5–7 meters long, which focus the beams.

The LHCb experiment [2] is one of the four large detectors at the Large Hadron Collider (LHC) accelerator at CERN, located at interaction point number 8 on the LHC ring. Its primary purpose is to search for new physics through studies of CP-violation and decays of heavy-flavour hadrons. Although LHCb was designed primarily for precision measurements in heavy-flavour physics, the experiment has demonstrated excellent capabilities in many other domains ranging from electroweak physics to heavy-ion and fixed-target physics. The LHCb Upgrade experiment has been designed with this wider physics programme in mind as a general purpose experiment covering the forward region. LHCb has been successfully operated from 2010 to 2018 during the LHC Run 1 (2010-2012) and Run 2 (2015-2018) data-taking periods with excellent performance, collecting a total of 9 fb^{-1} of proton-proton (pp) data, about 30 nb^{-1} of lead-lead and p -lead collisions and about 200 nb^{-1} of fixed target data.

2.2 The LHCb detector

The LHCb detector [2] is a single-arm forward spectrometer covering the pseudorapidity range $2 < \eta < 5$ (see subsection 3.2). Figure 2 shows the layout of the upgraded detector. The coordinate system of the LHCb detector

has the origin at the nominal pp interaction point, the z axis along the beam pointing towards the muon system, the y axis pointing vertically upward and the x axis defining a right-handed system.

The LHCb Upgrade I detector was designed to operate at a higher instantaneous luminosity based on the new trigger system. It consists of a new tracking system, a dipole magnet, and particle identification systems. The tracking system comprises an array of pixel silicon detectors surrounding the interaction region called vertex locator (VELO), the silicon-strip upstream tracker (UT) in front of the large-aperture dipole magnet, and three scintillating fibre tracker (SciFi) stations downstream of the magnet. All three subsystems were designed to comply with the 40 MHz readout architecture and to address the challenges associated with the increased luminosity.

The particle identification (PID) is provided by two ring imaging Cherenkov detectors (RICH1 and RICH2) using C₄F₁₀ and CF₄ gases as radiators, a shashlik-type electromagnetic calorimeter (ECAL), an iron-scintillator tile sampling hadronic calorimeter (HCAL), and four stations of muon chambers (M2–5) interleaved with iron shielding.

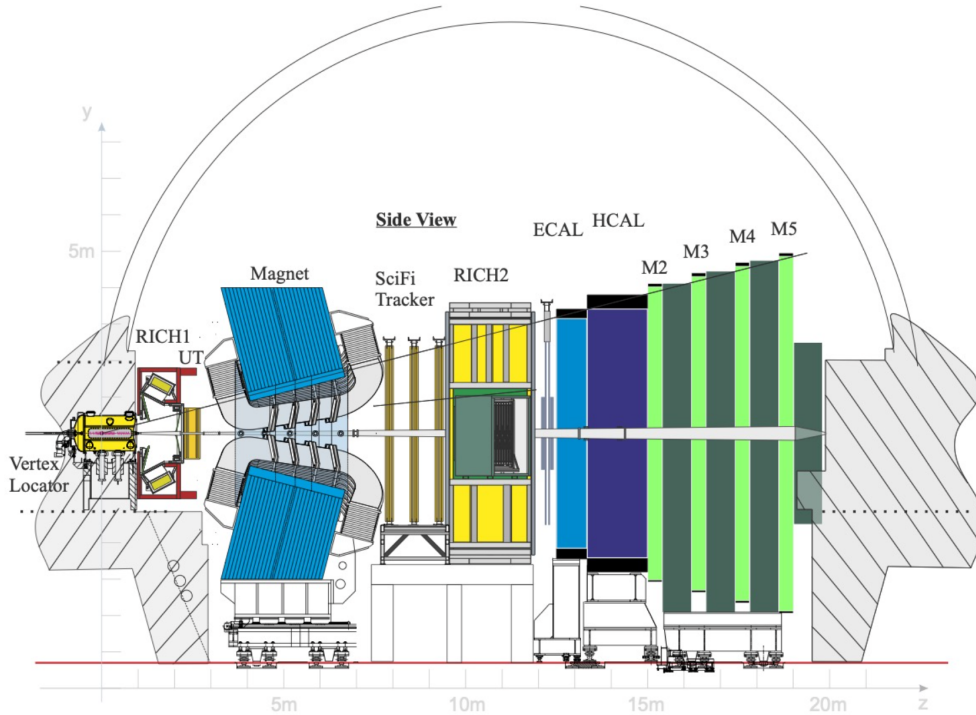


Figure 2: Layout of the LHCb experiment. [2]

2.3 Calorimeter system

The LHCb calorimeter system is particularly important for this study because the ECAL is used for photon detection, which is the main focus of this report. The main purpose of the LHCb calorimeter system [7], seen on Figure 3, is to trigger on electrons, photons and hadrons. It also provides energy and position measurements of the particles produced in their angular acceptance, which are used in offline event analysis. Furthermore, the electromagnetic calorimeter measures photons and neutral pions in association with the hadronic calorimeter.

The electromagnetic calorimeter (ECAL) [2] front surface is located at about 12.5 m from the interaction point. The square cell sizes for the outer, middle and inner regions are 121.2 mm, 60.6 mm, 40.4 mm side for outer, middle and inner regions, respectively, and scale with the distance from the beam-pipe in order to make the particle rate per cell roughly uniform. The outer dimensions of the ECAL match projectively those of the tracking system, $\theta_x < 300$ mrad and $\theta_y < 250$ mrad, while the inner angular acceptance of ECAL is limited to

$\theta_{x,y} > 25$ mrad around the beam pipe, where θ_x and θ_y are the polar angles in the xz and yz planes in the LHCb reference frame.

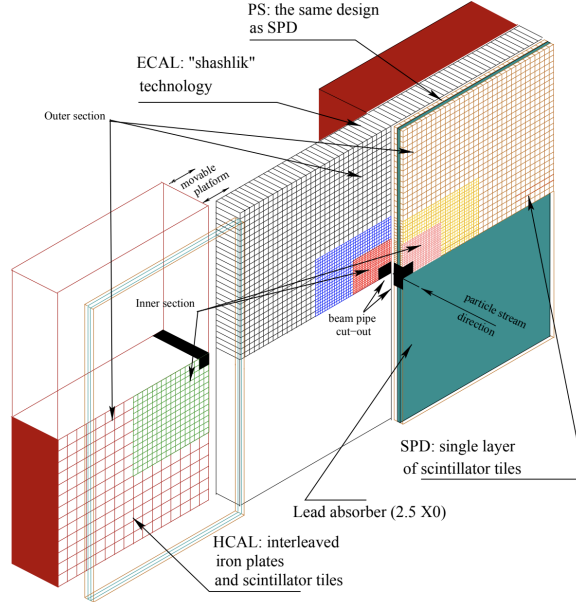


Figure 3: Layout of the calorimeter system. [8]

2.4 The fixed-target programme at LHCb: SMOG2

LHCb provides the unique opportunity at the LHC to operate in a fixed target mode, thanks to the System for Measuring Overlap with Gas (SMOG) [9]. Originally designed for precise luminosity measurements, SMOG allows the injection of noble gases such as argon or helium inside the primary LHC vacuum around the LHCb vertex detector (VELO). Since 2015, LHCb has started to exploit SMOG to perform physics runs, using special fills not devoted to pp physics, with different beam and target configurations, allowing unique production studies which are relevant to cosmic-ray and heavy-ion physics.

SMOG2 [10] is the next generation of the fixed-target system – a dedicated gas storage cell installed upstream of the VELO. It allows simultaneous operation with beam-beam and beam-gas collisions. The luminosity is up to 100 times higher than Run 2 SMOG programme. Furthermore, with SMOG2, a wide range of gases can be injected during Run 3, including hydrogen, deuterium, helium, neon, argon, xenon.

3. SIMULATION SETUP AND PROJECT STRATEGY

3.1 Simulation samples

In this section, we describe the simulation samples used in this project. The Monte Carlo production chain includes several steps: event generation, detector simulation, digitization, reconstruction, trigger, and sprucing.

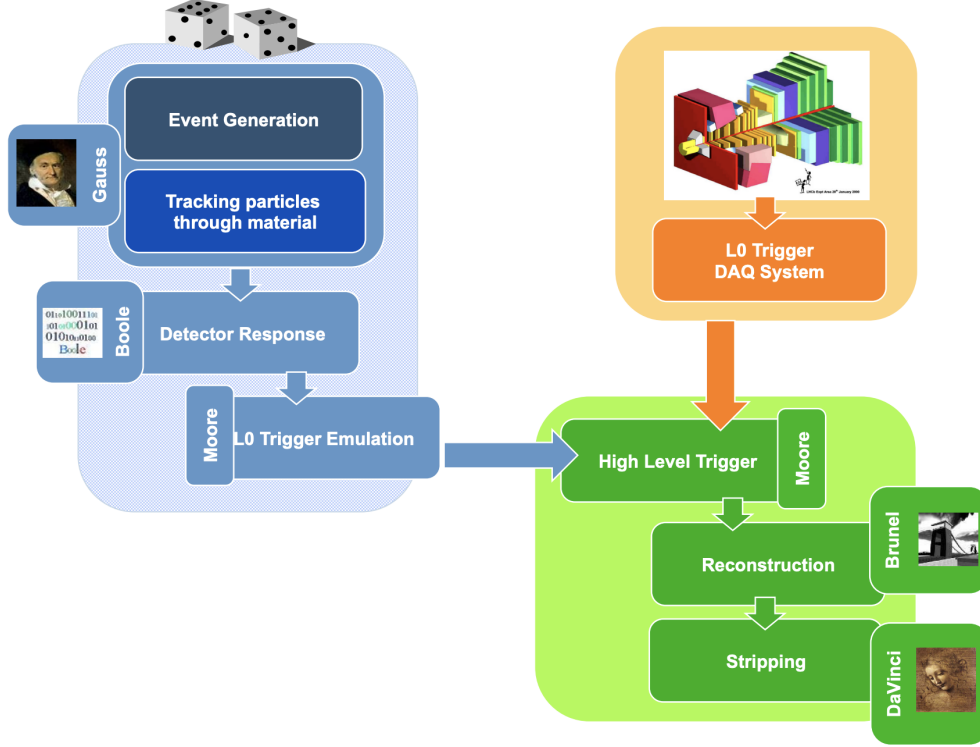


Figure 4: Outline of the LHCb data processing. On the left in blue the processing of Monte Carlo samples, the Data Taking in orange on the right. The Reconstruction and Stripping for both the MC and the data collected by the DAQ of the experiment are in green. [1]

The first two parts of the production are performed with GAUSS. It includes the following main components:

- **Event generation:** The signal particles are generated with PYTHIA8 [11]. Then, their decays are simulated using EVTGEN [12], while final-state radiation of muons is handled by PHOTOS [13]. Only events containing the required signal decay channel are kept. These signal events are then embedded into minimum-bias events, which are generated with EPOS generator.
- **Detector simulation:**
 - **Detailed simulation:** After the event generation, the passage of the generated particles through the LHCb detector is simulated using the GEANT4 [4] toolkit. This step models the interactions of the particles with the detector material and the corresponding detector response. In a typical physics event, more than 50% of CPU time is spent in the calorimeter [3], making it the natural starting point for developing a faster detector simulation.
 - **Fast Calo simulation:** After the event generation, the detector response can also be simulated using a fast-simulation approach. In this project, the calorimeter response is obtained from the calorimeter point-library [14], which provides pre-simulated detector responses for particles entering the calorimeter. This allows the main detector effects to be included with a significantly reduced computing cost compared to a full GEANT4-based detector simulation.

The simulation samples used in this project include both fast-simulation and detailed-simulation samples. They are produced for the 2024 fixed-target collision systems at $\sqrt{s_{NN}} = 70.9$ GeV. For all samples, the generated signal decay channel is $\chi_c \rightarrow J/\psi(\rightarrow \mu^+\mu^-)\gamma$. The fast-simulation samples contain about $\sim 10^6$ events for each collision system, while the detailed-simulation samples contain about $\sim 10^7$ events.

3.2 Variables of interest

The main goal of this project is validation of the photon performance in the decay chain $\chi_c \rightarrow (J/\psi \rightarrow \mu^+\mu^-)\gamma$, for the purpose of studying χ_c productions in SMOG2 collisions. Therefore, we are mainly interested in the photon variables and the χ_c mass. The photon variables we investigate are the variables related to photon kinematics and cluster information in ECAL. The kinematics variables are the photon momentum and its components (p , p_x , p_y , p_z), the angular information ($\phi = \text{atan2}(p_y, p_x)$, $\theta = \arccos(p_z/p)$ – derived, see [Appendix A](#)), and the pseudorapidity $\eta = -\ln \tan \frac{\theta}{2}$. Note that the pseudorapidity contains only angular information, which happens to coincide with the actual rapidity $y = \frac{1}{2} \ln \frac{E+p_z}{E-p_z}$ for the case of zero mass, therefore for photons.

3.3 Truth matching

For simulated samples, truth matching is used to identify reconstructed candidates that correspond to the intended generated decay chain. This confirmation is done by using the variables `TRUEID` and `MC_MOTHER_ID`. For the decay $\chi_c \rightarrow (J/\psi \rightarrow \mu^+\mu^-)\gamma$, we require both the reconstructed particle identities to match their truth IDs and the mother relations to reproduce the generated decay chain. The exact conditions used are given in [Appendix B](#). After applying the truth matching requirement, the size of simulation samples is significantly reduced. The simulation size decrease for experiments investigated in this report are given below in [Table 1](#). In our analyses, truth matching is applied for all plots until it is explicitly stated otherwise, in order to analyze the performance of the true photons coming from χ_c decay.

Table 1: Exact simulation size before and after truth matching.

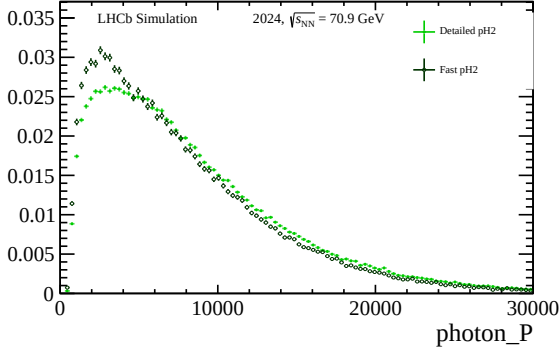
Collision system	Detailed simulation size	Size with truth matching	After/Before ratio
2024 $p\text{H}_2$	19445398	1169841	6.02%
2024 $pp\text{H}_2$	75326684	1037760	1.38%
2024 $p\text{Ar}$	39621866	1364486	3.44%
2024 $pp\text{Ar}$	51376796	605460	1.18%
2024 PbAr	16334557	75447	0.46%

4. VALIDATION RESULTS IN pH_2 COLLISIONS

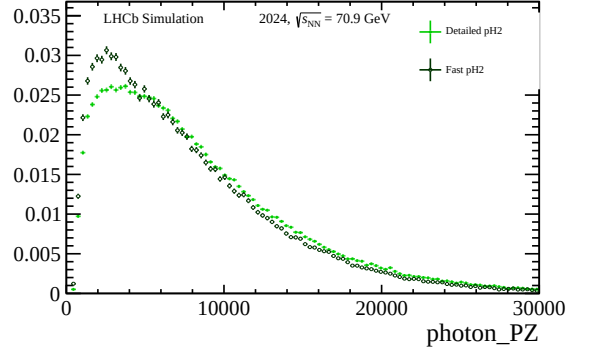
We start the investigation of the performance of the fast simulation in the simplest system, pH_2 collisions. In this collision system, we validate how the performance of the fast simulation depends on several relevant variables, including photon energy, or the calorimeter region that is studied.

4.1 Photon kinematics

Photon kinematics can be characterized by the photon momentum, rapidity, and the spherical angles θ and ϕ . In the comparison between detailed and fast simulations, several small discrepancies are observed across these kinematic distributions.



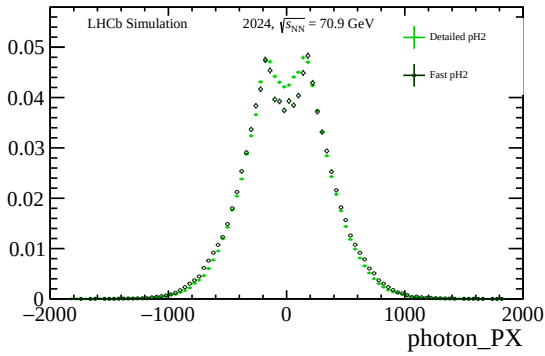
(a) `photon_P`



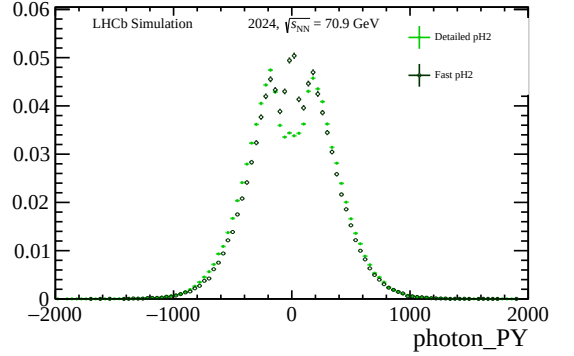
(b) `photon_PZ`

Figure 5: Comparison of detailed and fast simulation on `photon_P` and `photon_PZ` distributions in pH_2 collisions.

The discrepancies observed for `photon_P` and `photon_PZ` are very similar: the fast simulation overestimates the distribution significantly in the `photon_[P,PZ] ∈ [0, 6000]` MeV/c region. The performance in low momentum region is observed to be worse than the high momentum region.



(a) `photon_PX`



(b) `photon_PY`

Figure 6: Comparison of detailed and fast simulation on `photon_PX` and `photon_PY` distributions in pH_2 collisions.

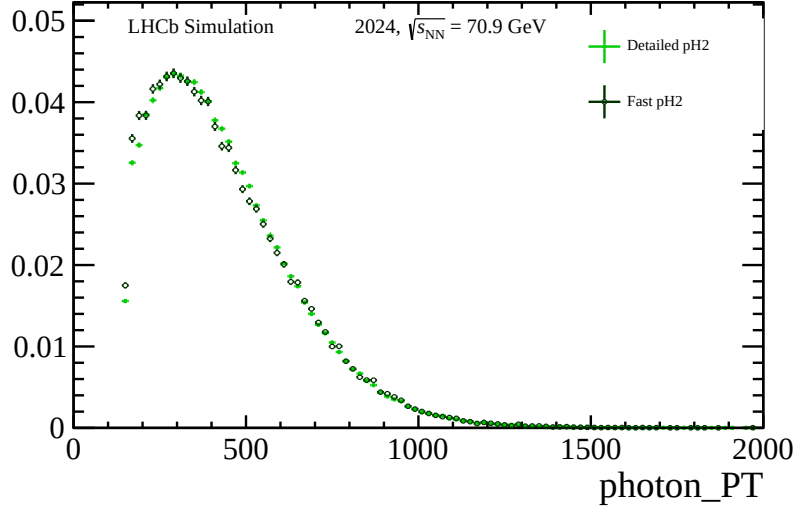


Figure 7: Comparison of detailed and fast simulation on `photon_PT` distribution in $p\text{Pb}$ collisions.

The distributions and the discrepancies are very similar for `photon_PX` and `photon_PY`. We see that in the range `photon_[PX,PY] ∈ [−200, 200]` MeV/c, the fast simulation underestimates for `photon_PX` and overestimates for `photon_PY`. However, when we observe the distribution of the transverse momentum $p_T = \sqrt{p_x^2 + p_y^2}$, we see that the over- and underestimations compensate each other mostly, resulting in a good fit for `photon_PT`.

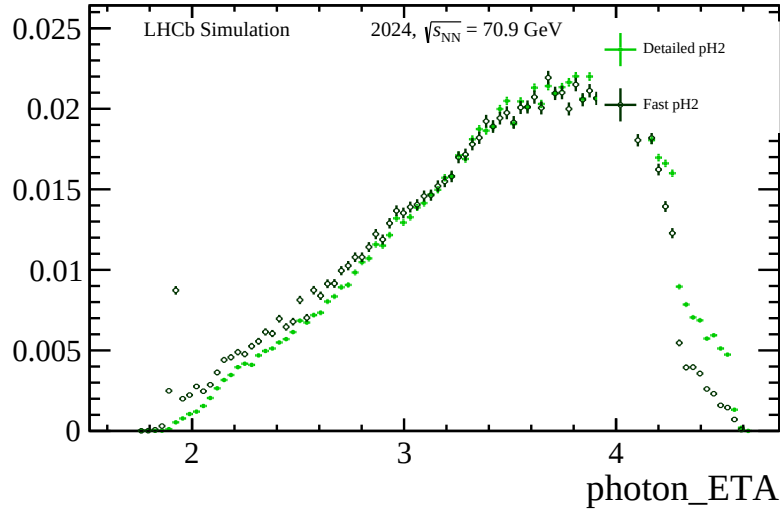
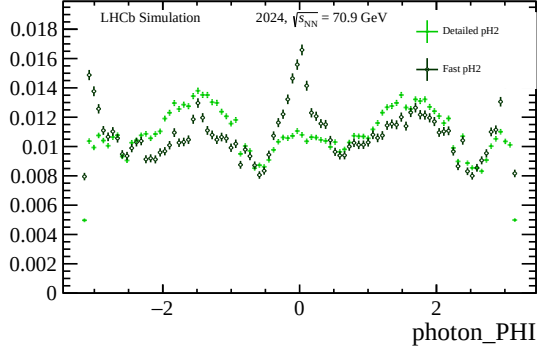
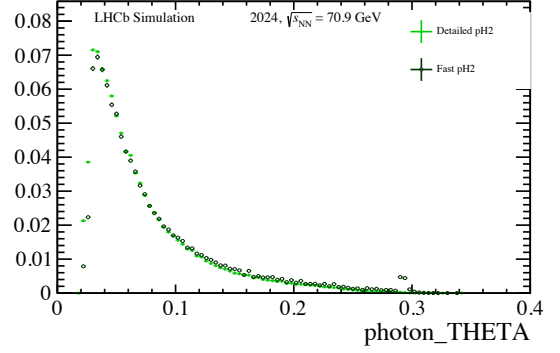


Figure 8: Comparison of detailed and fast simulation on `photon_ETA` distribution in $p\text{Pb}$ collisions.

The photon pseudorapidity, `photon_ETA`, exhibits similar profiles for the fast and detailed simulations, but there are regional differences. The fast simulation overestimates in the `photon_ETA < 3` range and underestimates in `photon_ETA > 4`.



(a) photon_PHI



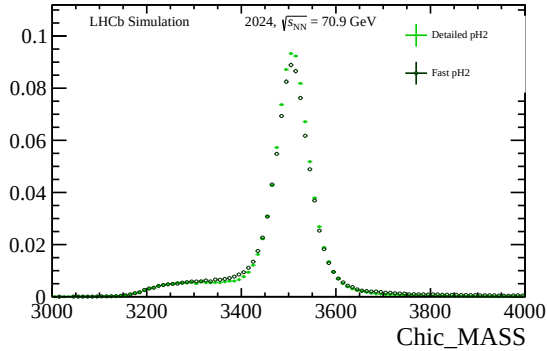
(b) photon_THETA

Figure 9: Comparison of detailed and fast simulation on `photon_PHI` and `photon_THETA` distributions in pH_2 collisions.

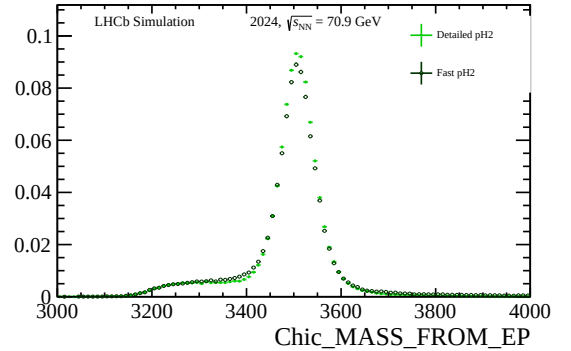
The `photon_PHI` distribution from the fast simulation has discrepancies all around the range, but the significant discrepancies are the overestimation around `photon_PHI` ≈ 0 and π . The θ angle distribution for photons is not provided in the tree of simulation samples. It is therefore computed from the photon momentum components, with the derivation given in [Appendix A](#). The distribution of this variable is one of the best estimated distributions by the fast simulation, however the overestimation at `photon_THETA` ≈ 0.3 is significant, and should be noted. This overestimation is present in different collision systems too (see [Figure 23](#)).

4.2 Investigation of χ_c mass

The reconstructed χ_c mass is one of the relevant observables for validating the fast simulation, since it will reflect the photon reconstruction resolution. The J/ψ is reconstructed from two well-reconstructed muon tracks, while the photon relies on the calorimeter response, which is precisely the part affected by the fast simulation. Therefore, any mismatch in the photon kinematics can directly propagate to the reconstructed χ_c invariant-mass shape.



(a) Chic_MASS



(b) Chic_MASS_FROM_EP

Figure 10: Comparison of detailed and fast simulation on `Chic_MASS` and `Chic_MASS_FROM_EP` distributions in pH_2 collisions.

We validate the performance of the fast simulation on the `Chic_MASS` variable, and the `Chic_MASS_FROM_EP` variable derived from `Chic_[ENERGY,P]` ([Appendix A](#)). We see that the two distributions have the same profiles, and the fast simulation performs mostly well in both cases, except for the slight overestimate near `Chic_MASS` ≈ 3400 MeV/ c^2 . We investigate several selections affecting the `Chic_MASS` distribution.

In order to validate the performance of the fast simulation in different photon energy ranges, we perform cuts at `photon_ENERGY` < 6 GeV and `photon_ENERGY` > 20 GeV.

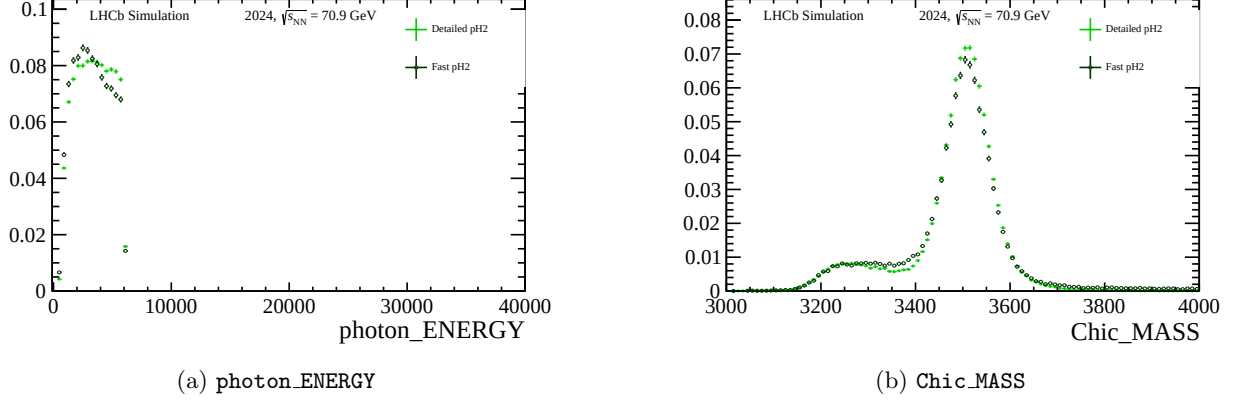


Figure 11: Comparison of detailed and fast simulation on `Chic_MASS` distribution in pH_2 collisions with low energy cut `photon_ENERGY` < 6 GeV, and the corresponding `photon_ENERGY` distribution.

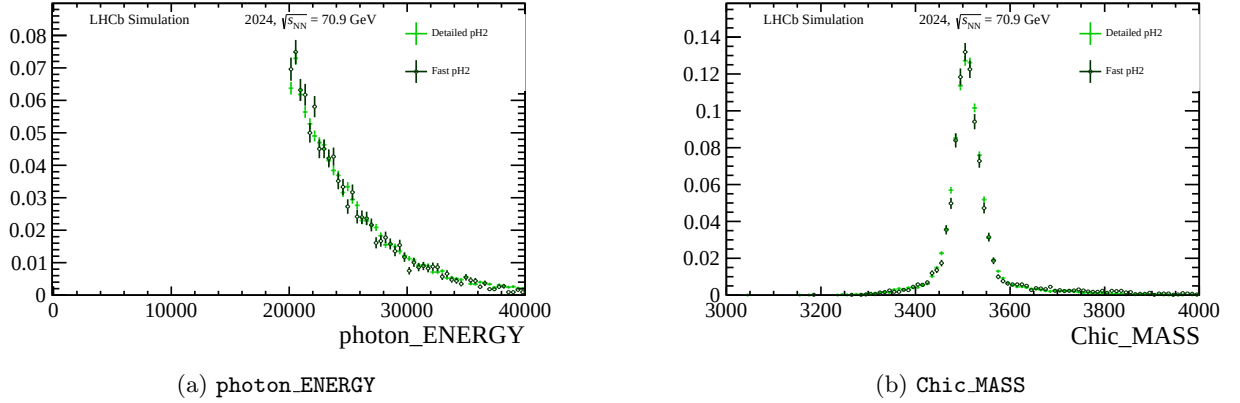


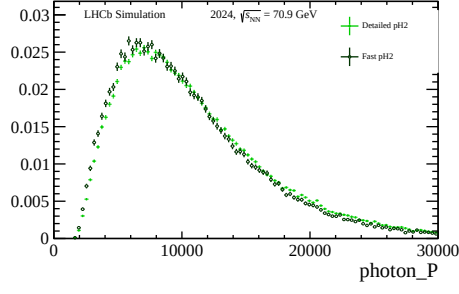
Figure 12: Comparison of detailed and fast simulation on `Chic_MASS` distribution in pH_2 collisions with high energy cut `photon_ENERGY` > 20 GeV, and the corresponding `photon_ENERGY` distribution.

From these distributions, we see that most of the discrepancies in the total distribution is caused by the low momentum samples, and that the fast simulation performs better for high photon energies.

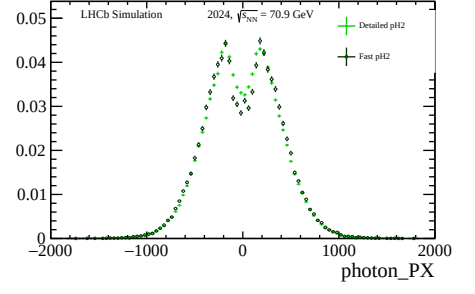
4.3 Performance validation by calorimeter region

It is also important to validate photon performance between the detailed and fast simulations in different calorimeter regions. The LHCb ECAL is divided into three regions (inner, middle and outer) with different cell sizes in order to adapt the detector granularity to the particle density. The region which is closer to the beam will have higher particle density, thus higher granularity of ECAL. Therefore, one needs to validate the photon performance among these three regions.

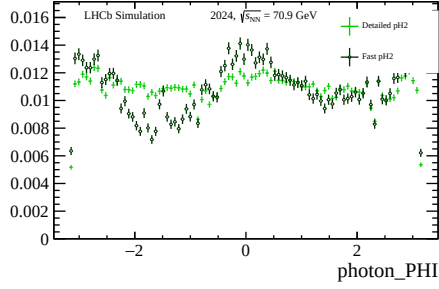
In order to validate the performance of the fast simulation across different calorimeter regions, we use the variable `photon_CaloNeutralArea` to perform the cuts. Note that variables of similar profiles are not duplicated, namely `photon_[P, PZ]` and `photon_[PX, PY]`.



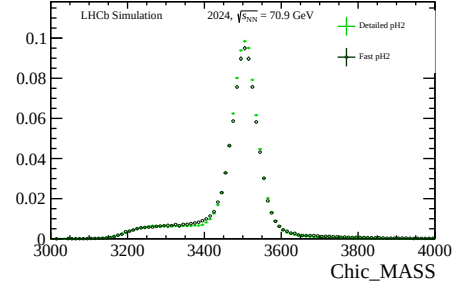
(a) photon_P



(b) photon_PX

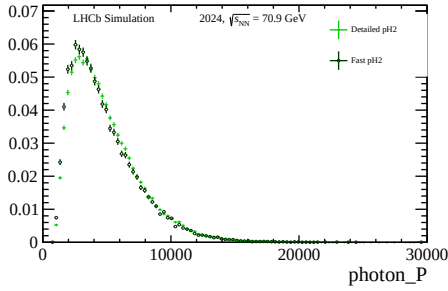


(c) photon_PHI

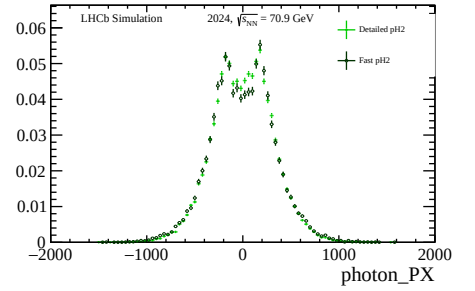


(d) Chic_MASS

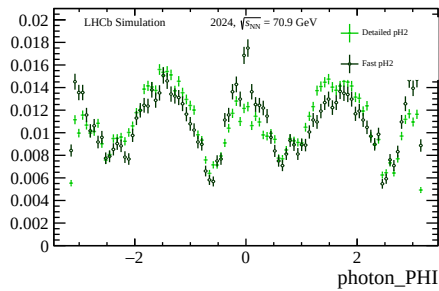
Figure 13: Comparison of detailed and fast simulation on photon_P, photon_PX, photon_PHI and Chic_MASS distributions in pH_2 collisions, in the inner calorimeter region with photon_CaloNeutralArea = 3.



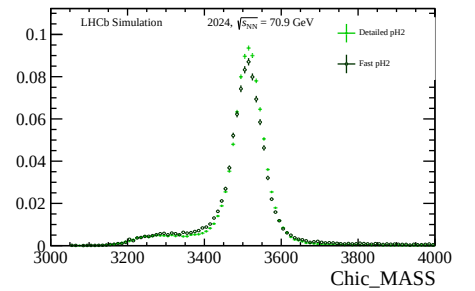
(a) photon_P



(b) photon_PX

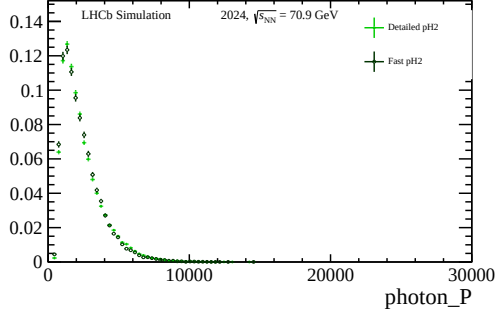


(c) photon_PHI

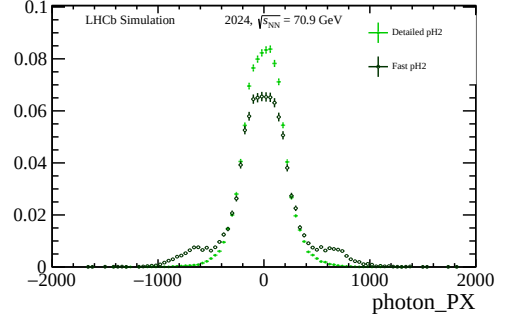


(d) Chic_MASS

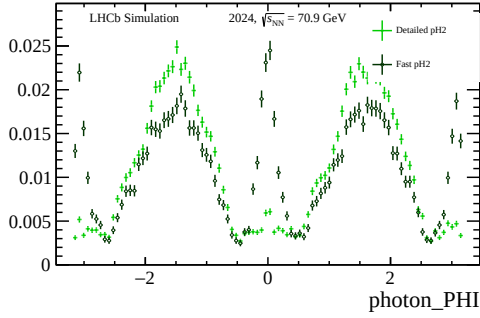
Figure 14: Comparison of detailed and fast simulation on photon_P, photon_PX, photon_PHI and Chic_MASS distributions in pH_2 collisions, in the middle calorimeter region with photon_CaloNeutralArea = 2.



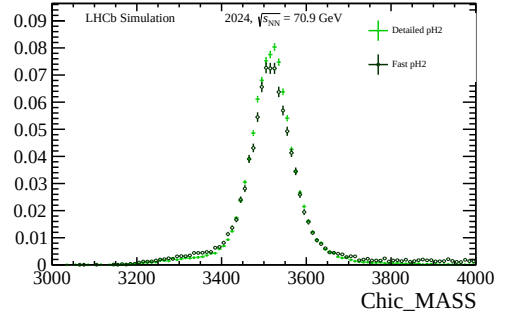
(a) photon_P



(b) photon_PX



(c) photon_PHI



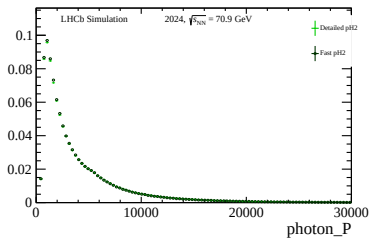
(d) Chic_MASS

Figure 15: Comparison of detailed and fast simulation on `photon_P`, `photon_PX`, `photon_PHI` and `Chic_MASS` distributions in pH_2 collisions, in the outer calorimeter region with `photon_CaloNeutralArea` = 1.

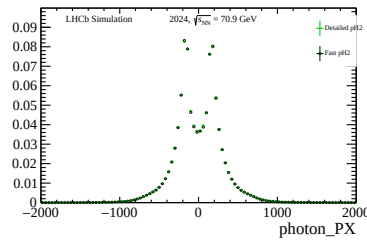
We see that the fast simulation shows better fit on `photon_P` when the regions are investigated separately. For `photon_PX`, the worst performance is on the outer section, while for `photon_PHI` the middle section has better performance compared to the other two. The `Chic_MASS` distribution is not affected by the change of the calorimeter region.

4.4 Effect of truth-matching

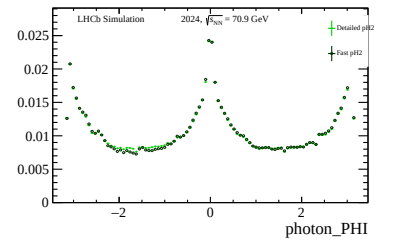
To study the impact of the truth-matching requirement, the truth-matching condition is removed. In this way, one can validate the photon performance between fast and detailed simulation among all reconstructed photons. The selection is therefore changed to the inclusive `1==1` requirement, as described in [Appendix B](#).



(a) photon_P



(b) photon_PX



(c) photon_PHI

Figure 16: Comparison of detailed and fast simulation on `photon_P`, `photon_PX` and `photon_PHI` distributions in pH_2 collisions, without the truth matching condition.

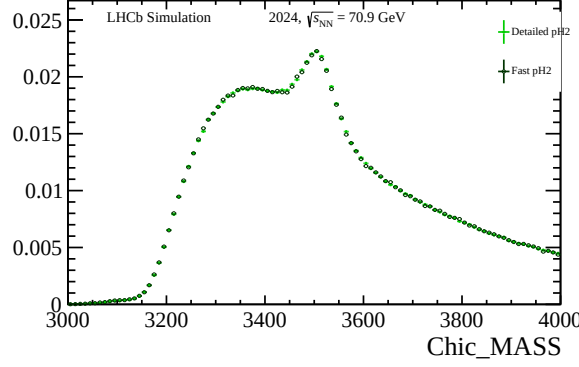


Figure 17: Comparison of detailed and fast simulation on `Chic_MASS` distribution in $p\text{Pb}$ collisions, without the truth matching condition.

From Figure 16 and Figure 17, one can observe that the photon kinematic distributions change significantly when the truth-matching requirement is removed. This is expected, since photons originating from the χ_c decay (with truth matching) are expected to have different kinematic properties from other background photons (without truth matching).

The most prominent difference is observed in the `Chic_MASS` distribution. When the truth-matching requirement is removed, the reconstructed $J/\psi\gamma$ invariant-mass spectrum is no longer dominated by photons originating from the χ_c decay, but contains a large contribution from background photons. As a result, the shape of the `Chic_MASS` distribution changes significantly: the χ_c signal region around 3500 MeV/c^2 becomes largely buried under the combinatorial background.

However, when comparing the fast and detailed simulations without truth matching requirement, one can see that the fast simulation performance is significantly improved, with only small differences observed.

5. EXTENSION TO OTHER SMOG2 COLLISIONS

In this section, we extend the comparisons to different collision systems with more complex nuclear environments, namely ppH_2 , pAr , $ppAr$, and $PbAr$ collisions.

5.1 Detailed simulation comparison across systems

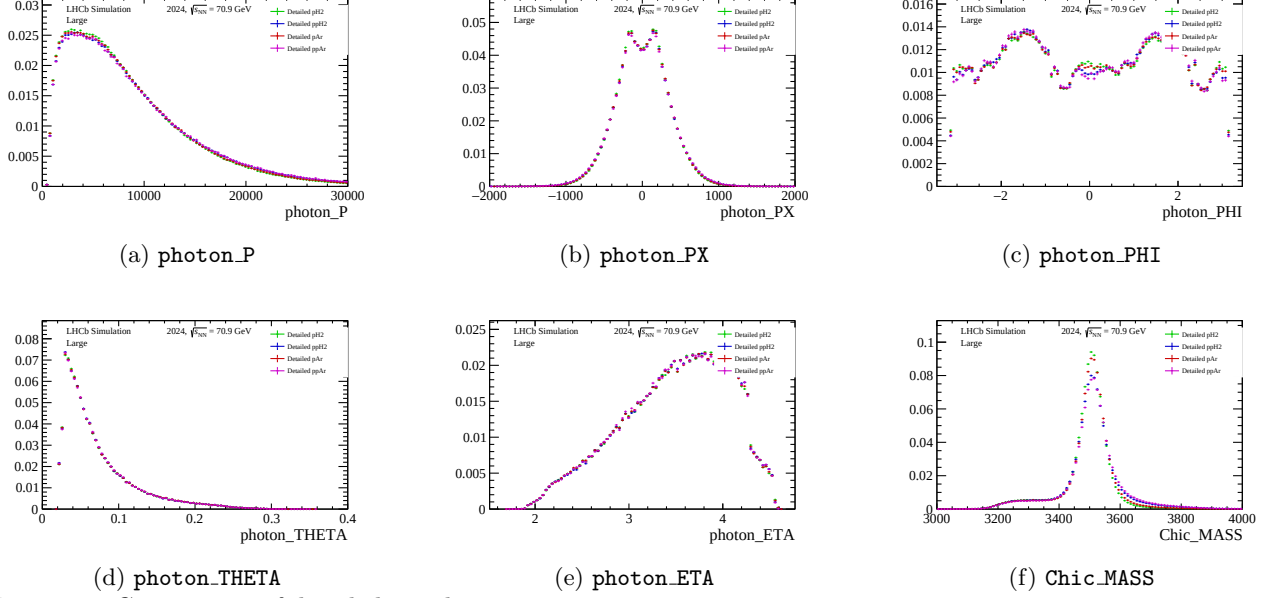
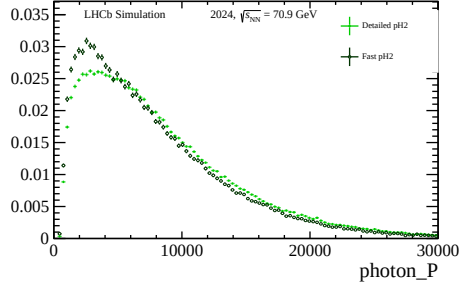


Figure 18: Comparison of detailed simulation on photon_P , photon_PX , photon_PHI , photon_THETA , photon_ETA and Chic_MASS distributions in pH_2 , ppH_2 , pAr and $ppAr$ collisions.

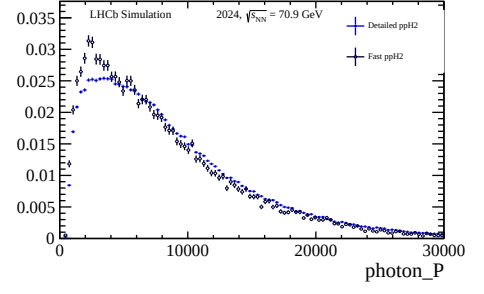
As shown in Figure 18, there are no significant differences in the distributions of photon kinematics and χ_c mass in the ppH_2 , pAr and $ppAr$ collision systems compared to pH_2 .

5.2 Fast vs Detailed simulation across systems

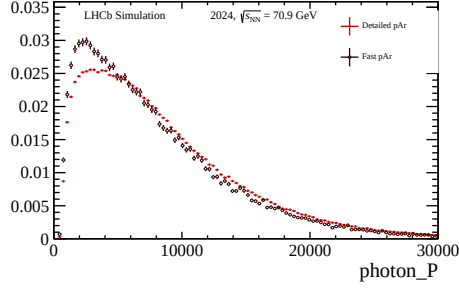
We compare the results from the fast and detailed simulations across different collision systems for different variables.



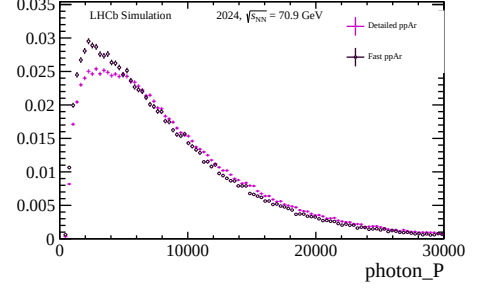
(a) pH_2



(b) ppH_2



(c) pAr



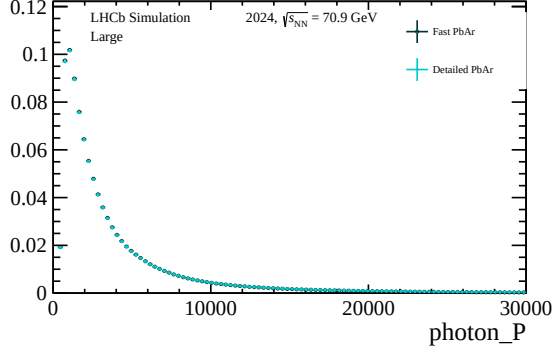
(d) $ppAr$

Figure 19: Comparison of detailed and fast simulation on `photon_P` distribution in pH_2 , ppH_2 , pAr and $ppAr$ collisions.

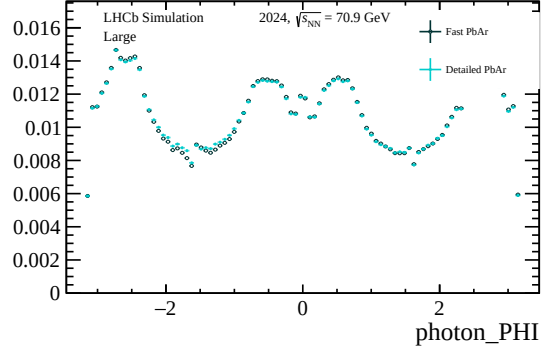
As an example, we investigate the `photon_P` distribution, shown on Figure 19. (The corresponding distributions for the other variables are provided in Appendix C.) We observe that the discrepancies across different collision systems have similar patterns between fast and detailed simulations.

5.3 Cross-check in PbAr collisions

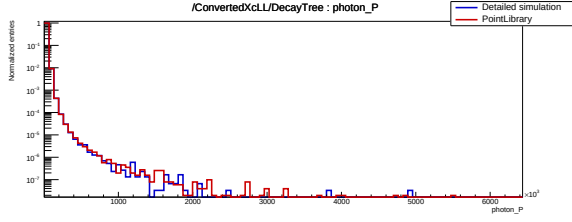
We reported our comparison studies in p -nucleus collision systems to the LHCb fast simulation experts. The photon performance comparison was also performed on the PbAr collision without applying the truth-matching requirement. Figure 20 compares the results obtained with our validation procedure to those provided by the fast-simulation development team. The two sets of distributions are found to be consistent, with no visible discrepancy observed.



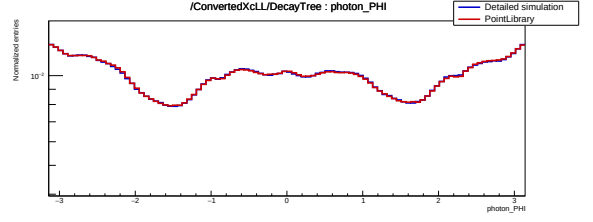
(a) **photon_P**



(b) **photon_PHI**



(c) **photon_P** cross-check



(d) **photon_PHI** cross-check

Figure 20: Comparison of detailed and fast simulation on **photon_P** and **photon_PHI** distributions in PbAr collisions. Results in this project are shown in the top plots, while the cross-check plots from the fast-simulation development team are shown in the bottom.

In these plots, we do not apply truth matching. Thus, as we expect from our comparisons for pH_2 without truth matching, we get a good fit of the fast simulation with the detailed simulations, and similar distributions to the cross-check plots.

6. CONCLUSION

In this report, the performance of the LHCb fast calorimeter simulation was studied in fixed-target collisions at $\sqrt{s_{NN}} = 70.9$ GeV by using $\chi_c \rightarrow (J/\psi \rightarrow \mu^+\mu^-)\gamma$ decay. The validation was performed by comparing fast-simulation samples with detailed GEANT4-based simulation samples for several 2024 collision systems: pH_2 , ppH_2 , pAr , $ppAr$, and $PbAr$ collisions. Since the reconstruction of the χ_c signal depends directly on the photon reconstructed in the electromagnetic calorimeter, the comparison focused mainly on photon kinematic variables, photon angular variables, calorimeter-region dependence, truth-matching effects, and the reconstructed χ_c mass.

For the baseline pH_2 system, the fast simulation reproduces the general shape of most photon distributions, but several discrepancies are observed. In particular, the photon momentum and longitudinal momentum distributions show an overestimation by the fast simulation at low momentum, and an underestimation at higher momentum. The transverse momentum distribution is better reproduced, partly because differences in the p_x and p_y components compensate each other. The photon pseudorapidity and azimuthal-angle distributions also show local differences, while the polar-angle distribution is among the best described variables, apart from a visible excess around $\theta \simeq 0.3$. The reconstructed χ_c mass is generally well reproduced, with only a small overestimation near the signal peak.

The study of different photon-energy regions shows that the largest discrepancies mainly come from low-energy photons, while the agreement between fast and detailed simulation improves for high photon energies. This suggests that the fast calorimeter simulation performs better in the high-energy region and that the low-energy photon response should be investigated more carefully. The comparison by calorimeter region shows that the agreement can improve when the inner, middle, and outer calorimeter regions are treated separately. This indicates that part of the discrepancy in the inclusive distributions may come from region-dependent effects in the calorimeter response.

The role of truth matching was also investigated. When the truth-matching requirement is removed, the distributions change significantly. This effect is particularly visible in the reconstructed χ_c mass distribution, where the χ_c mass peak becomes buried under a large combinatorial background from photons. This behavior is expected, since photons originating from the χ_c decay and background photon candidates have different kinematic distributions. At the same time, the agreement between fast and detailed simulation becomes much better for most variables. This shows that the validation result depends strongly on the selected candidate sample, and that truth-matched and non-truth-matched comparisons test different aspects of the simulation and reconstruction chain.

The comparison was then extended to ppH_2 , pAr , and $ppAr$ collisions. The detailed-simulation distributions are broadly similar across these systems. In the fast-versus-detailed comparisons, the observed discrepancies follow similar patterns in the different collision systems. This suggests that the main differences between fast and detailed simulation are not specific to one fixed-target configuration, but are rather related to common features of the photon and calorimeter modeling.

Finally, the $PbAr$ sample was studied without truth matching in order to compare with studies from the fast-simulation development team. In this case, the fast simulation shows good agreement with the detailed simulation, consistently with the previous non-truth-matched pH_2 comparison.

Overall, the fast calorimeter simulation gives a reasonable description of the main photon and χ_c mass distributions in LHCb fixed-target χ_c samples. However, non-negligible discrepancies remain, especially for low-energy photons, some angular variables, and truth-matched samples. These effects should be studied further before using the fast simulation in precision analyses of χ_c production in SMOG2 data. Possible future work includes a more quantitative comparison using ratio plots and goodness-of-fit metrics, a more detailed study of the calorimeter-region dependence, and an investigation of the origin of the differences observed after truth matching.

This validation work provides valuable feedback for the development of the LHCb fast simulation, and supports its future use in analyses involving photons in complex SMOG2 collision environments.

APPENDIX A. DERIVED VARIABLES

- $\text{photon_THETA} = \arccos(\text{photon_PZ}/\text{photon_P})$
- $\text{Chic_MASS_FROM_EP} = \sqrt{\text{Chic_ENERGY}^2 - \text{Chic_P}^2}$
- $\text{photon_E_reco_E_true} = \text{photon_ENERGY} - \text{photon_TRUEENERGY}$

APPENDIX B. SELECTION PARAMETERS

Various selection methods used in the analyses done in this report are highlighted below. The selection filters are fed in the string format to the plotting script.

- Without truth matching: "1==1"
- With truth matching:

```
Truth = "Jpsi_TRUEID==Jpsi_ID && photon_TRUEID==photon_ID && Chic_TRUEID==Chic_ID  
&& mup_TRUEID==mup_ID && mum_TRUEID==mum_ID  
&& Jpsi_MC_MOTHER_ID==Chic_ID && photon_MC_MOTHER_ID==Chic_ID  
&& mup_MC_MOTHER_ID==Jpsi_ID && mum_MC_MOTHER_ID==Jpsi_ID"
```

- Low photon energy cut with truth matching:

```
"photon_ENERGY<6000 && " + Truth
```

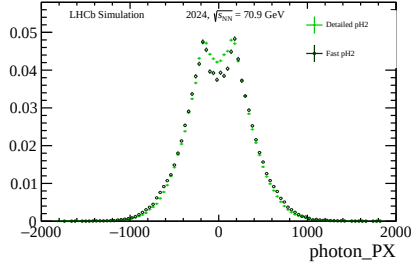
- High photon energy cut with truth matching:

```
"photon_ENERGY>20000 && " + Truth
```

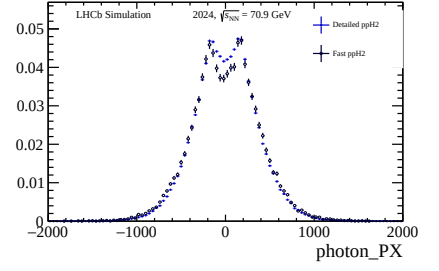
- Calorimeter region (outer, middle, inner) with truth matching:

```
"photon_CaloNeutralArea==[1, 2, 3] && " + Truth
```

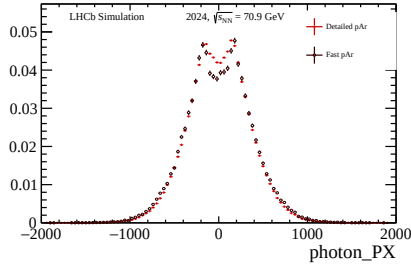
APPENDIX C. VALIDATION OF THE FAST SIMULATION IN DIFFERENT SYSTEMS – ADDITIONAL VARIABLES



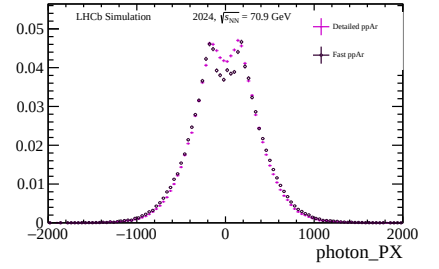
(a) pH_2



(b) ppH_2

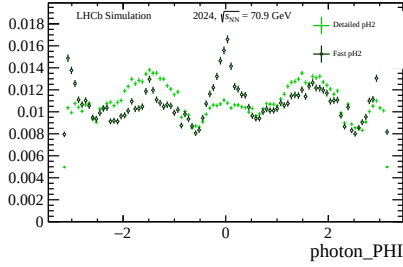


(c) pAr

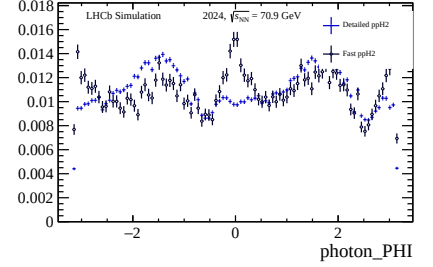


(d) $ppAr$

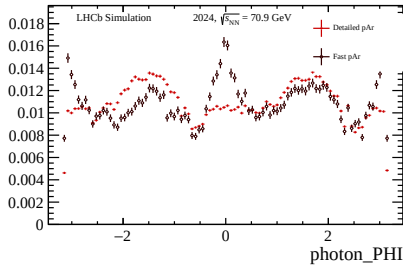
Figure 21: Comparison of detailed and fast simulation on `photon_PX` distribution in pH_2 , ppH_2 , pAr and $ppAr$ collisions.



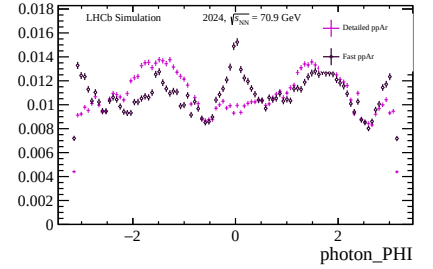
(a) pH_2



(b) ppH_2

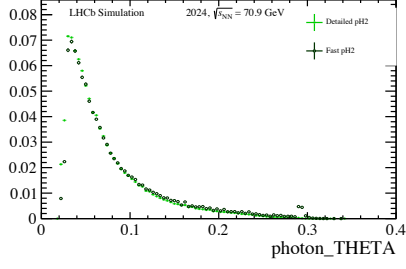


(c) pAr

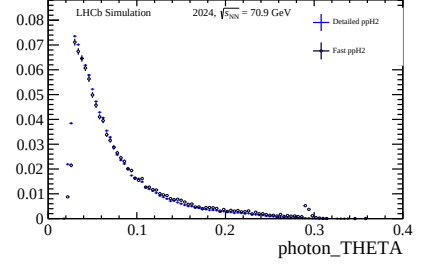


(d) $ppAr$

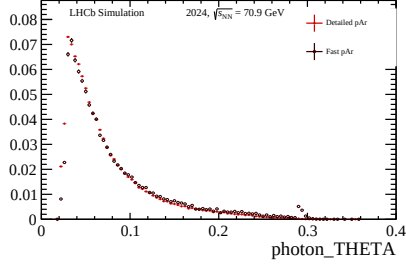
Figure 22: Comparison of detailed and fast simulation on `photon_PHI` distribution in pH_2 , ppH_2 , pAr and $ppAr$ collisions.



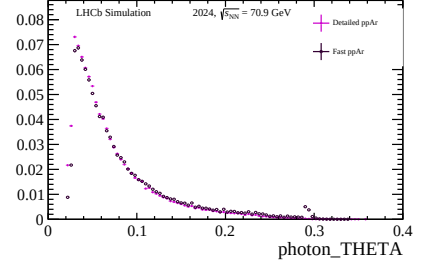
(a) pH_2



(b) ppH_2

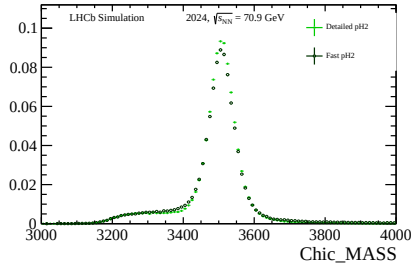


(c) pAr

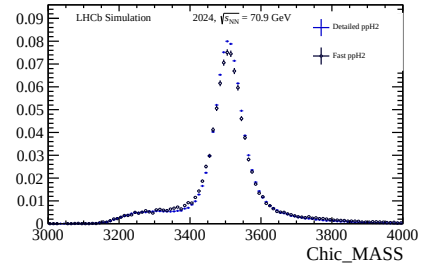


(d) $ppAr$

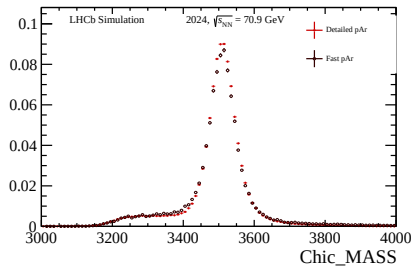
Figure 23: Comparison of detailed and fast simulation on `photon_THETA` distribution in pH_2 , ppH_2 , pAr and $ppAr$ collisions.



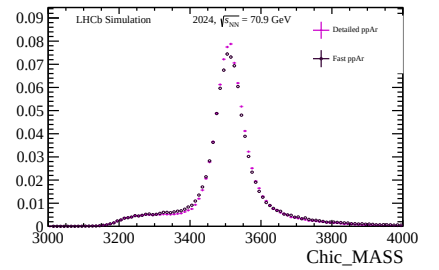
(a) pH_2



(b) ppH_2



(c) pAr



(d) $ppAr$

Figure 24: Comparison of detailed and fast simulation on `Chic_MASS` distribution in pH_2 , ppH_2 , pAr and $ppAr$ collisions.

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